

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** Complaint ease: Online Registrations And Management Systems
- **Team Members:** Individual Project

Name: T.Sree Akshitha

2. Project Overview

Purpose

The Online Complaint Registration and Management System is a web-based platform developed using the MERN Stack that enables users to register complaints online, track complaint status, communicate with assigned agents, and receive updates regarding complaint resolution.

The system aims to improve transparency, reduce manual paperwork, ensure timely complaint resolution, and provide an efficient complaint management process for users, agents, and administrators.

Features

- User Registration & Login
- JWT Authentication
- Role-Based Access (User, Agent, Admin)
- Complaint Registration
- Complaint Status Tracking
- Agent Assignment
- User-Agent Chat System
- Feedback Management
- Admin Dashboard
- Agent Dashboard
- Complaint History
- Email Notifications
- Responsive User Interface

3. Architecture

Frontend Architecture (React.js)

The frontend is developed using React.js with a component-based architecture.

Frontend Technologies

- React.js
- HTML5
- CSS3
- JavaScript
- React Router DOM
- Axios
- Bootstrap
- Context API

Frontend Modules

- Home Page
- Login Page
- Signup Page
- User Dashboard
- Agent Dashboard
- Admin Dashboard
- Complaint Form
- Complaint Status Tracking
- Chat Window
- Feedback Page

Backend Architecture (Node.js & Express.js)

The backend follows MVC Architecture.

Backend Technologies

- Node.js
- Express.js
- JWT Authentication
- bcryptjs
- express-validator
- dotenv
- cors
- nodemailer
- mongoose

Backend Modules

- Authentication Controller
- Complaint Controller
- Admin Controller
- Agent Controller
- Chat Controller
- Feedback Controller

Database Architecture (MongoDB)

MongoDB Atlas is used as the cloud database.

Collections

Users

```
{
  "name": "string",
  "email": "string",
  "password": "string",
  "phone": "number",
  "role": "USER | AGENT | ADMIN"
}
```

Complaints

```
{
  "title": "string",
  "description": "string",
  "category": "string",
  "status": "Pending | Assigned | In Progress | Resolved",
  "userId": "ObjectId",
  "agentId": "ObjectId"
}
```

Feedback

```
{
  "rating": "number",
  "comment": "string",
  "complaintId": "ObjectId"
}
```

Messages

```
{
  "senderId": "ObjectId",
  "receiverId": "ObjectId",
  "complaintId": "ObjectId",
}
```

```
"message": "string"
}
```

4. Setup Instructions

Install:

- Node.js
- MongoDB Atlas
- VS Code
- Git

Frontend Setup

```
cd frontend
npm install
npm run dev
```

Backend Setup

```
cd backend
npm install
npm start
```

Environment Variables

PORT=5000

MONGO_URI=your_mongodb_connection_string

JWT_SECRET=your_secret_key

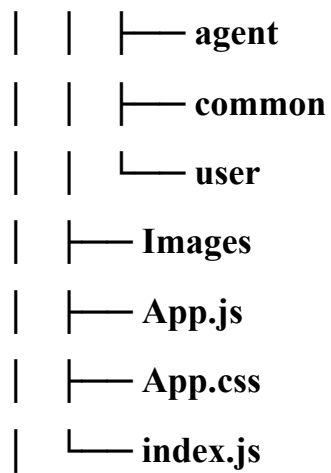
EMAIL_USER=your_email

EMAIL_PASS=your_email_password

5. Folder Structure

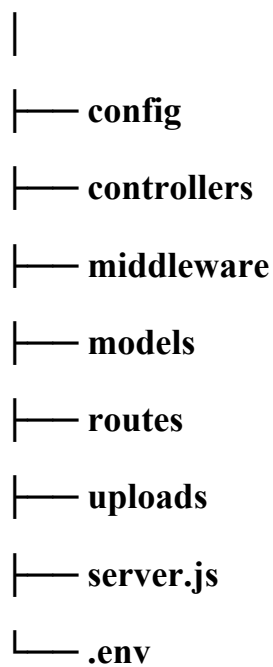
Frontend frontend

```
|
|
|— public
|— src
|  |— components
|  |  |— admin
```



Backend Structure

backend



6. API Documentation

Authentication APIs

Register User

Login User

POST /api/auth/login

Complaint APIs

Create Complaint
POST /api/complaints

Get Complaints

GET /api/complaints

Update Complaint

PUT /api/complaints/:id

Delete Complaint

DELETE /api/complaints/:id

Admin APIs

Assign Agent

PUT /api/admin/assign-agent/:id

View Users

GET /api/admin/users

View Agents

GET /api/admin/agents

7. API Documentation

Authentication is implemented using JWT.

Authentication Flow

```
graph TD
    A[User Login] --> B[JWT Token Generated]
    B --> C[Token Stored in Browser]
    C --> D[Protected Routes Accessed]
```

Security Features

- JWT Authentication
- bcrypt Password Encryption
- Role-Based Authorization
- Protected APIs

- Helmet Security
- CORS Protection

8. User Interface

UI Modules

- Home Page
- Login Page
- Signup Page
- User Dashboard
- Agent Dashboard
- Admin Dashboard
- Complaint Status Page
- Chat Window
- Feedback Page

UI Design

- Responsive Layout
- Bootstrap Styling
- Dashboard Cards
- Sidebar Navigation
- Toast Notifications
- Mobile Friendly Design

9. Testing

Testing Strategy

- API Testing using Postman
- Frontend Testing using Browser Developer Tools
- MongoDB Database Testing

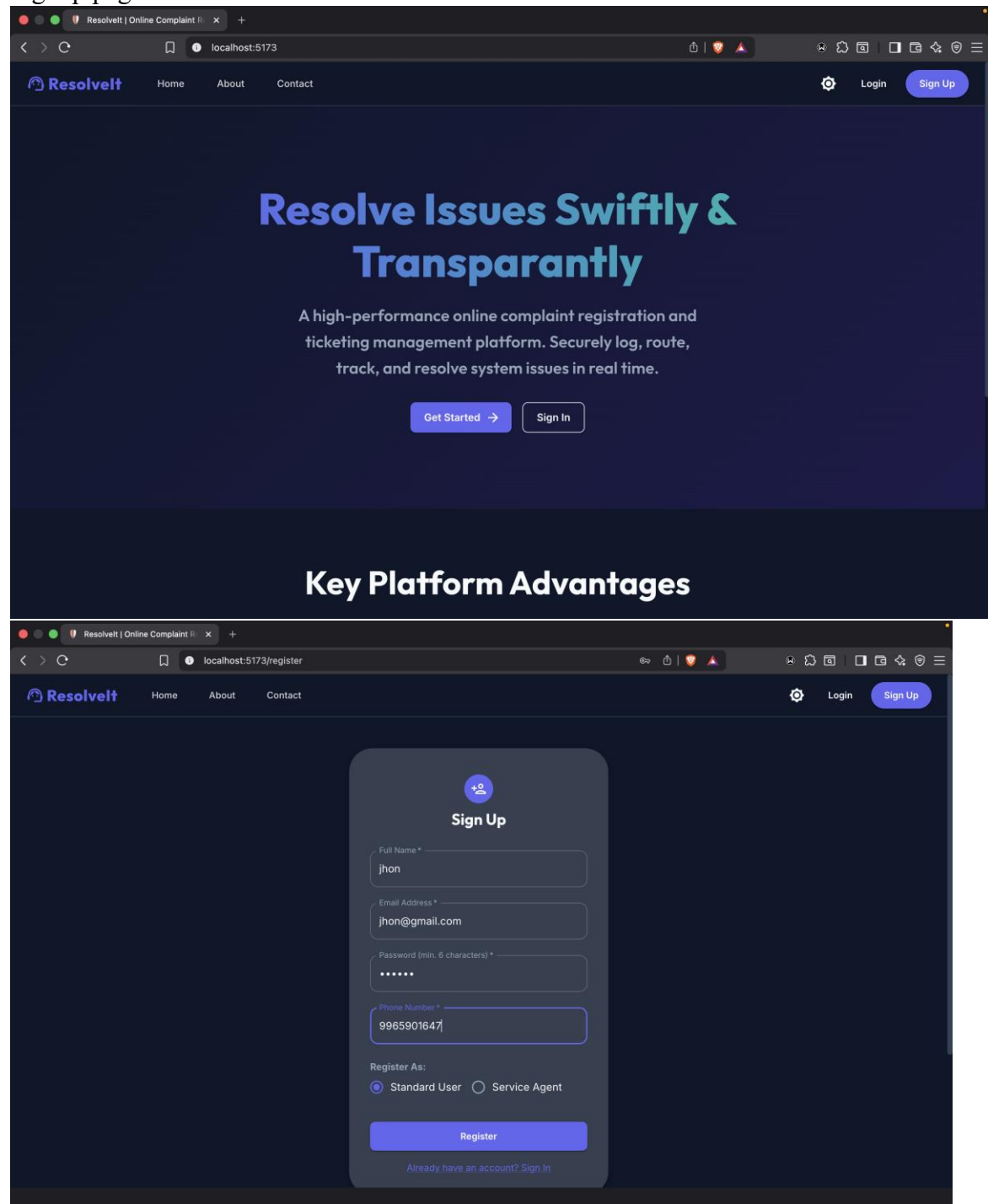
Tested Modules

- User Authentication
- Complaint CRUD Operations
- Agent Assignment
- Status Tracking
- Feedback Module
- Role-Based Access Control

10.Screenshots or Demo

Screenshots

Signup page:



The image shows two screenshots of a web application. The top screenshot is the homepage, and the bottom screenshot is the signup page.

Homepage:

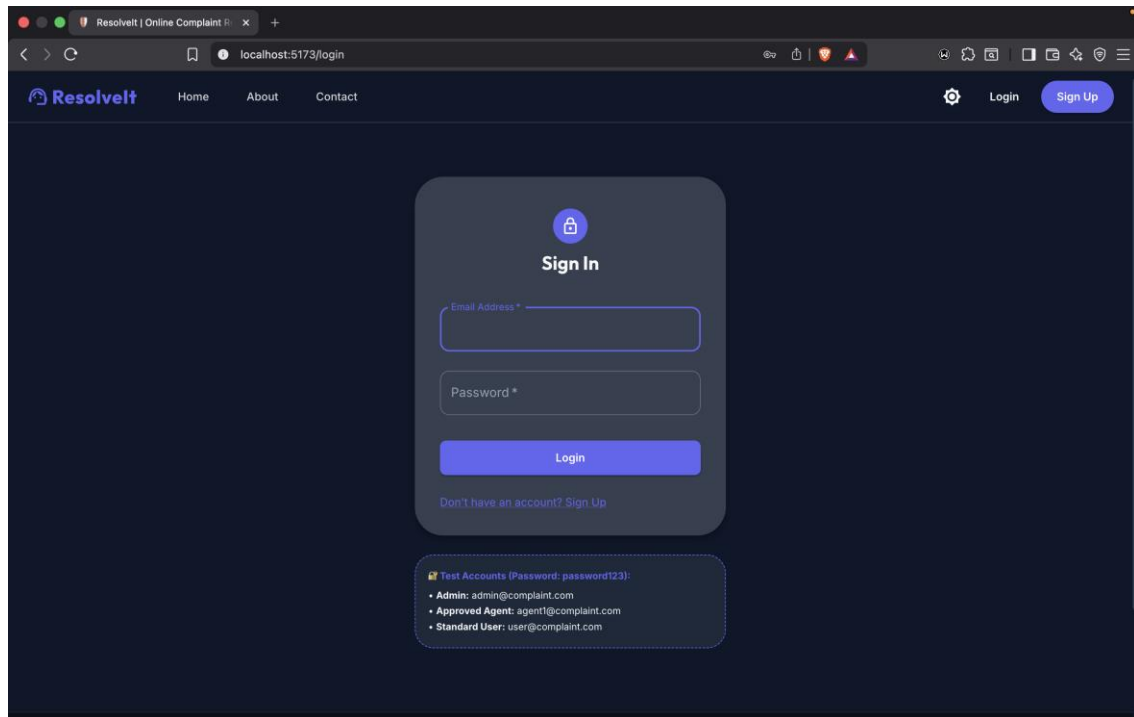
- Browser: Resolvelt | Online Complaint R... x +
- URL: localhost:5173
- Navigation: Home, About, Contact, Login, Sign Up
- Header: Resolvelt
- Main Content:
 - # Resolve Issues Swiftly & Transparently
 - A high-performance online complaint registration and ticketing management platform. Securely log, route, track, and resolve system issues in real time.
 - [Get Started →](#) [Sign In](#)
- Footer: Key Platform Advantages

Signup Page:

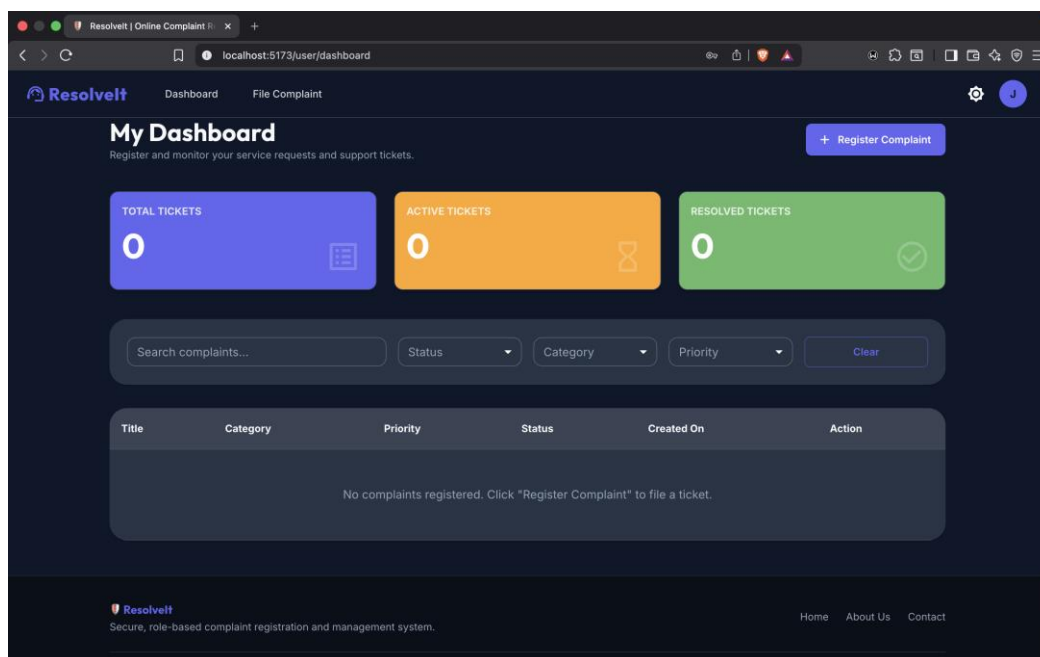
- Browser: Resolvelt | Online Complaint R... x +
- URL: localhost:5173/register
- Navigation: Home, About, Contact, Login, Sign Up
- Header: Resolvelt
- Form:
 - ## Sign Up
 - Full Name *
 - Email Address *
 - Password (min. 6 characters) *
 - Phone Number *
 - Register As:

☒ Standard User ☐ Service Agent
 - [Register](#)
 - Already have an account? [Sign In](#)

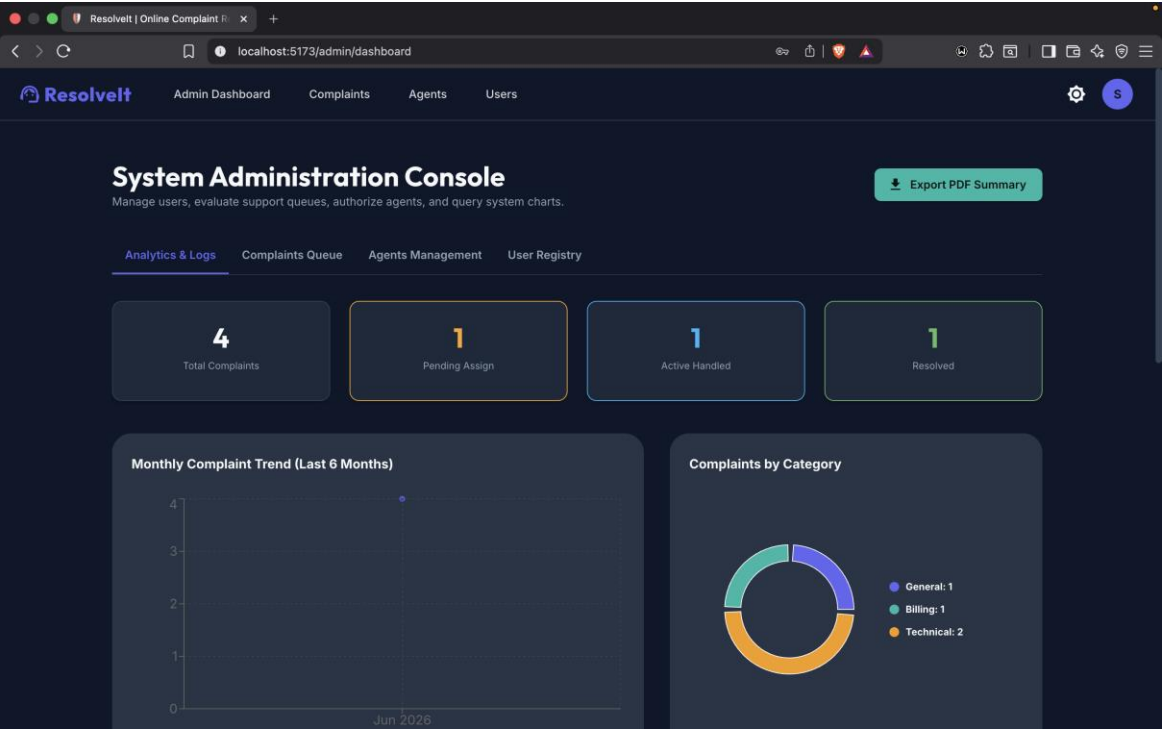
Login page:



User dashboard:



Admin Dashboard:



ResolveIt | Online Complaint R | x +

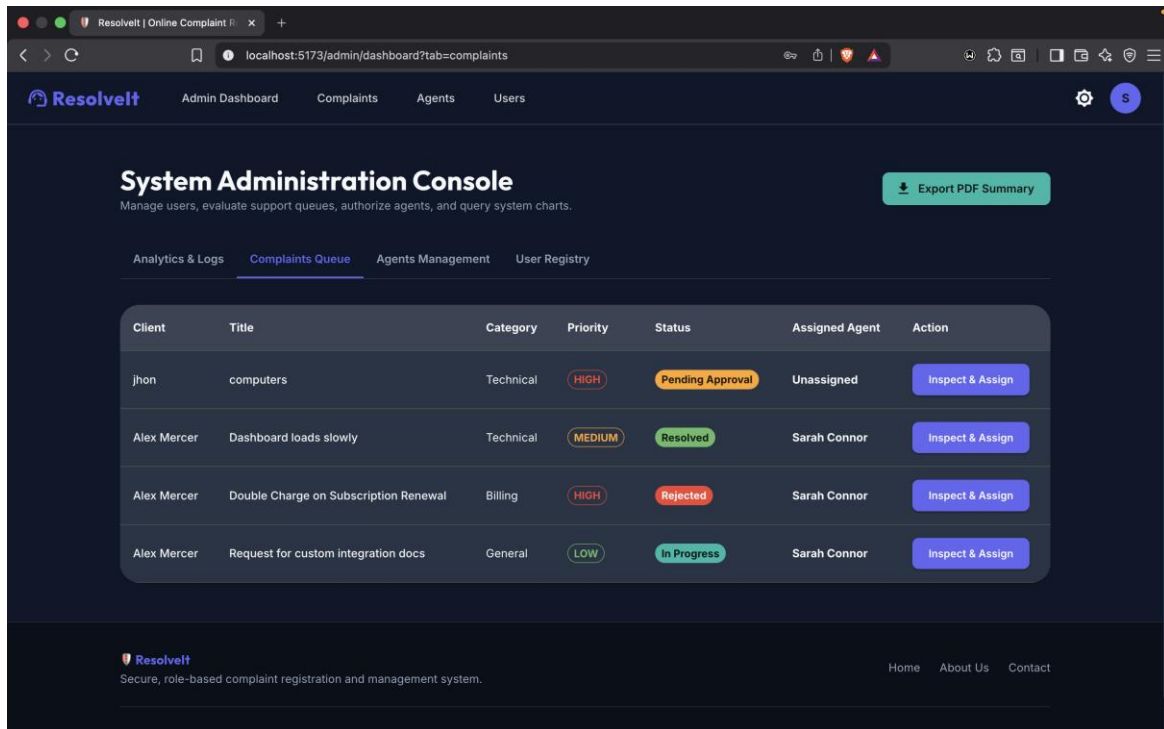
localhost:5173/admin/dashboard

Admin Dashboard Complaints Agents Users

Audit Logs (Recent Activity)

User	Role	Action	Details	IP Address	Timestamp
System Admin	ADMIN	LOGIN	User logged in successfully	::1	18/06/2026, 17:42:47
jhon	USER	COMPLAINT_CREATED	Complaint ID: 6a33dfd054df2272ffaaa875	::1	18/06/2026, 17:38:48
jhon	USER	REGISTER	User registered with role: USER	::1	18/06/2026, 17:32:44
Sarah Connor	AGENT	COMPLAINT_STATUS_UPDATED	Updated complaint 6a32f692839716c72725b852 status to IN_PROGRESS	::1	18/06/2026, 01:11:23
Sarah Connor	AGENT	COMPLAINT_STATUS_UPDATED	Updated complaint 6a32f692839716c72725b84b status to REJECTED	::1	18/06/2026, 01:11:07
Sarah Connor	AGENT	COMPLAINT_STATUS_UPDATED	Updated complaint 6a32f692839716c72725b84f status to RESOLVED	::1	18/06/2026, 01:10:52
Sarah Connor	AGENT	LOGIN	User logged in successfully	::1	18/06/2026, 01:10:22
System Admin	ADMIN	COMPLAINT_STATUS_UPDATED	Updated complaint 6a32f692839716c72725b852 status to ASSIGNED	::1	18/06/2026, 01:08:41
System Admin	ADMIN	COMPLAINT_ASSIGNED	Assigned complaint 6a32f692839716c72725b852 to Agent Sarah Connor	::1	18/06/2026, 01:08:40
System Admin	ADMIN	COMPLAINT_STATUS_UPDATED	Updated complaint 6a32f692839716c72725b84b status to ASSIGNED	::1	18/06/2026, 01:08:21

Complaint status page:



Include:

- Signup Page
- Login Page
- User Dashboard
- Admin Dashboard
- Complaint Status Page

Demo

GitHub Repository Link: <https://github.com/sreeakshitha11/VIP-C2-complaint-registration-system.git>

10. Known Issues

- Large file uploads may take additional time.
- Real-time chat performance depends on network connectivity.

- Email notifications may experience SMTP delays.

11. Future Enhancements

- Real-Time Chat using Socket.io
- SMS Notifications
- AI-Based Complaint Categorization
- Analytics Dashboard
- Mobile Application
- Push Notifications
- Complaint Priority Prediction
- Multi-Language Support

This version is customized specifically for your **SmartBridge Online Complaint Registration and Management System** and follows the format of the sample document you uploaded.